Open EFT for Cosmology & Gravity

Toshifumi Noumi (UTokyo-Komaba)

Mainly based on arXiv:2412.21136

w/ Pak Hang Chris Lau (Great Bay U) and Kanji Nishii (Keio U).

+ several previous / on-going works.

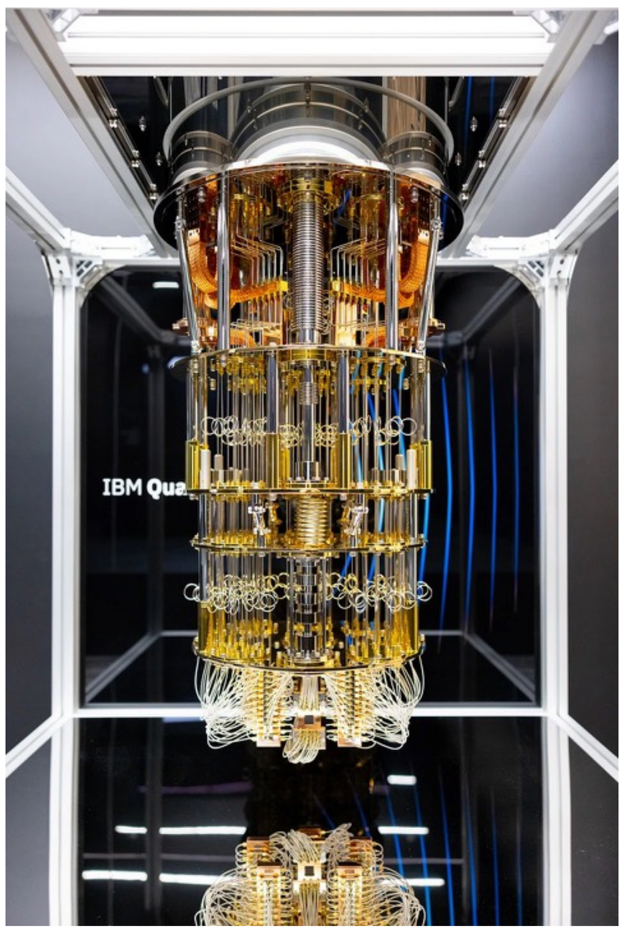
June 18th 2026 @ Theoretical Developments in Gravitational Wave Physics (YITP)

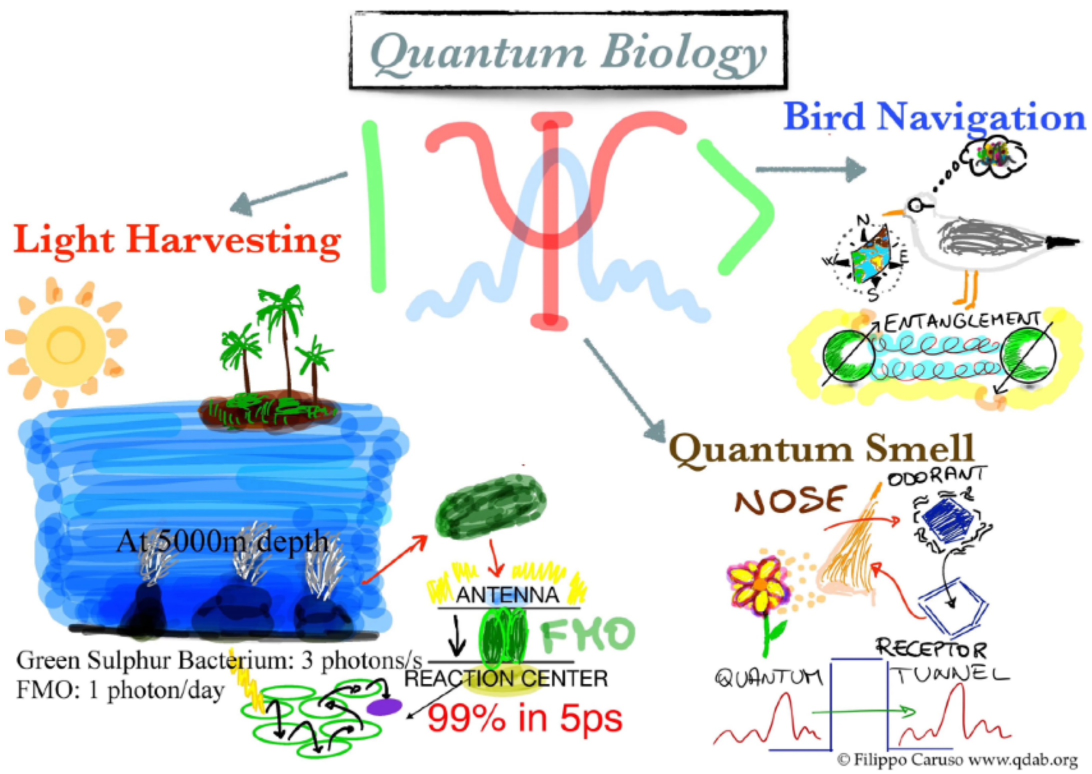
Introduction

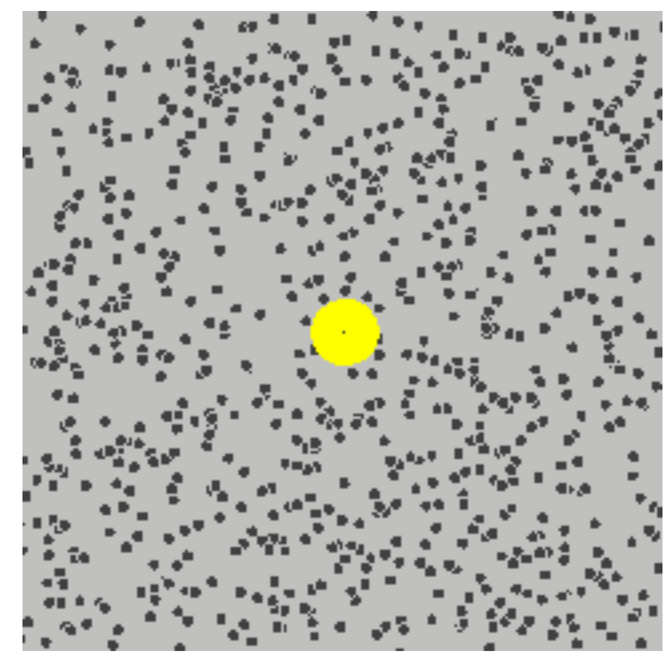
Open systems in nature

open system: a system coupled to surrounding environment

※ Energy, charge, information etc are exchanged with environment.



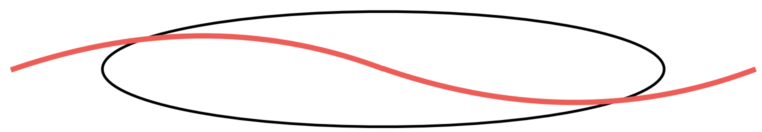




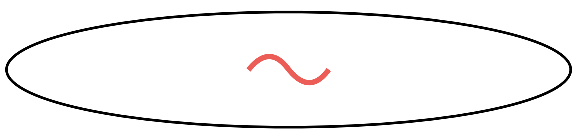
Our systems are always surrounded by an environment!

Why open systems in cosmology? (1/2)

Case 1: Cosmological horizon gives separation of scales / regions.



superhorizon

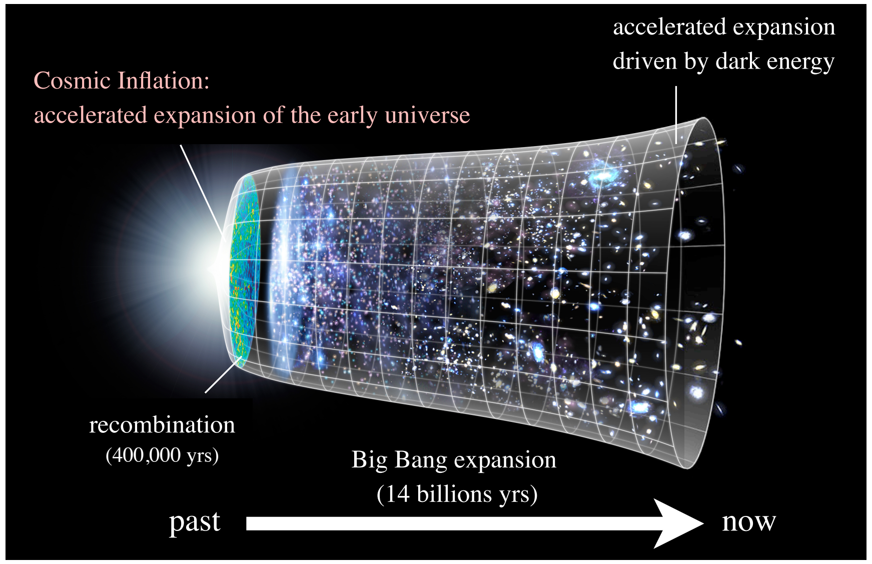


subhorizon

If we focus on dynamics of superhorizon / subhorizon modes,

subhorizon / superhorizon modes are identified with environment.

Case 2: Decoherence & residual coherence of primordial fluctuations



Entanglement between subsystems is useful to quantify quantumness.

Why open systems in cosmology? (2/2)

Case 3: Coupling to environment bath

ex. Our universe is filled with fluids

- EFT of Large Scale Structure = EFT of dissipative hydro
- If we consider some dof coupled to the fluids in the universe, the fluids play a role of the environment.

ex. Warm / dissipative inflation: $\ddot{\phi} + (3H+\gamma)\dot{\phi} + V'(\phi) = 0.$

- Dissipation slows the rolling inflaton

cf. quantum gravity corrections vs flatness of potential

It would also be great if cosmology offers new inspirations useful to understand other open quantum systems in nature!

It would be important to develop an EFT approach

for such studies of open system perspectives in cosmology.

Based on this motivation, I was working on EFT of open systems,

- Extension of EFT of inflation to open system [Hongo-Kim-**TN**-Ota ’18],

※ Recently, [Salcedo-Colas-Pajer ’24] named it *Open EFT of inflation* and studied phenomenology of primordial non-Gaussisnities.

- Coset construction for SSB in open systems [Hongo-Kim-**TN**-Ota ’19].

Also, more recently,

- Gravitational EFT for dissipative open systems [Lau-Nishii-**TN** ’24].

and several related works in progress.

In this talk, I discuss **how to incorporate dynamical gravity to open EFT**

with a short remark on application to tidal Love number & cosmic inflation.

[Mainly based on Lau-Nishii-**TN** '24]

See, e.g., [Salcedo et al 25, Christodoulidis-Gong '25, Kaplanek-Mylova-Tolley '25, '26, Abe-Nishii '26]

for recent related developments.

Contents

Dissipative scalar: an illustrative example

Consider a dissipative scalar ϕ coupled to a white noise ξ:

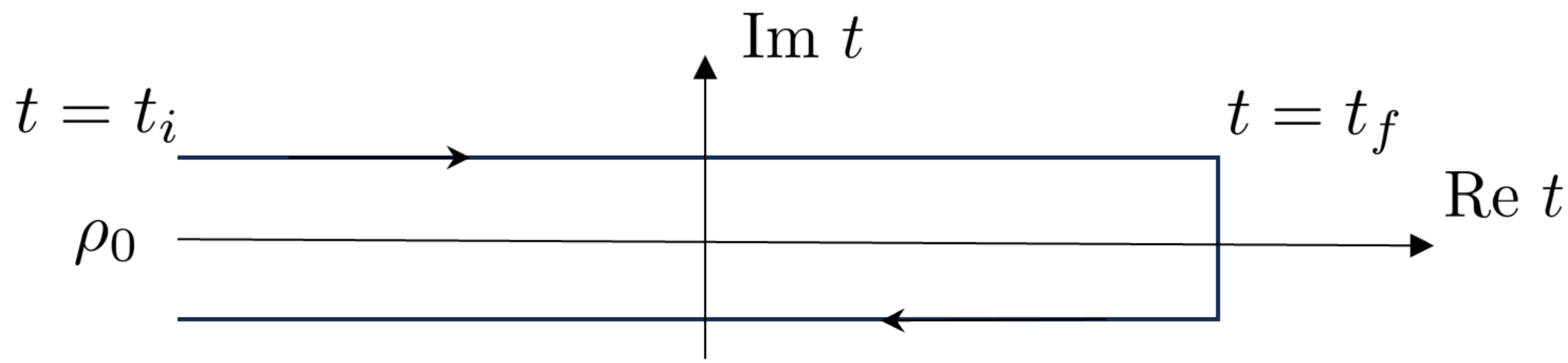
$$\ddot{\phi} + \gamma\dot{\phi} + V'(\phi) = \xi \quad \text{with} \quad \langle\, \xi(x)\, \xi(x')\, \rangle = A\, \delta^{(4)}(x - x').$$

※ $A = 2\gamma T$ holds for a thermal bath of temperature T.

The Schwinger-Keldysh formalism is a path-integral framework useful to describe such Langevin-type dynamics of open systems.

Schwinger-Keldysh description of the full sector

Schwinger-Keldysh / in-in formalism (1/2)



A fundamental quantity is the *closed-time-path* generating functional

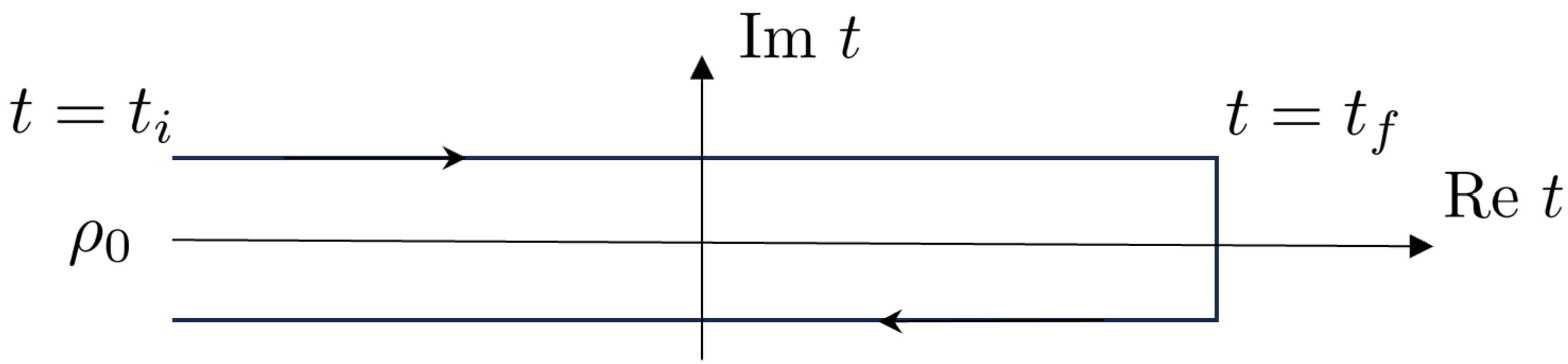
$W[J_1, J_2]$ defined by $e^{iW[J_1,J_2]} = \mathrm{Tr}\left[U_{J_1}(t_i, t_f)\,\rho_0\,U_{J_2}(t_f, t_i)^\dagger\right]$.

- ρ_0 : initial density matrix of the full sector

- $U_J(t_f, t_i)$: (unitary) time-evolution operator w/source J

※ Two types of sources J_1, J_2 are introduced.

Schwinger-Keldysh / in-in formalism (2/2)



The corresponding path-integral expression reads

$$e^{iW[J_1,J_2]} = \int D\Phi_1\, D\Phi_2\, \exp\left(iS[\Phi_1; J_1] - iS[\Phi_2; J_2]\right).$$

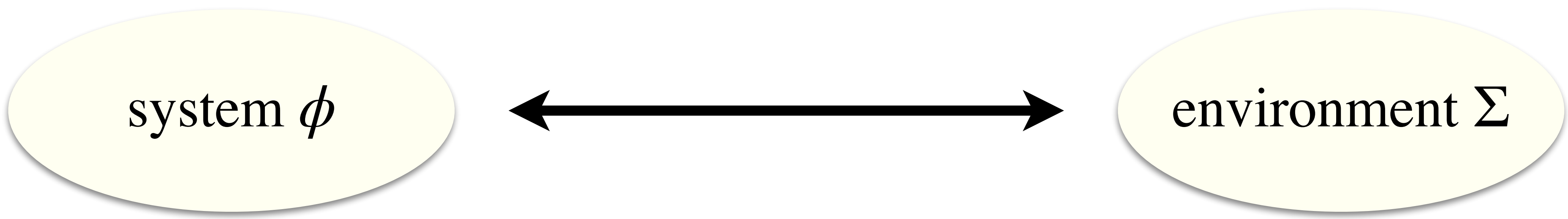
- Two sets of (Φ, J) are introduced.

- Boundary conditions at $t = t_f$: $\Phi_1(t_f) = \Phi_2(t_f)$

- Boundary conditions at $t = t_i$ is determined by the initial density matrix ρ_0.

Schwinger-Keldysh EFT of the system sector

SKEFT of the system sector (1/3)



Suppose that the unitary time-evolution of the full sector is described by the action $S[\phi, \Sigma]$, where Σ denotes environment dof collectively.

The SK effective action $S_{\text{eff}}[\phi_1, \phi_2]$ of ϕ is given by integrating Σ out:

$$\int D\phi_1 \, D\phi_2 \, \exp iS_{\text{eff}}[\phi_1, \phi_2] = \int D\phi_1 \, D\phi_2 D\Sigma_1 \, D\Sigma_2 \, \exp\left(iS[\phi_1, \Sigma_1] - iS[\phi_2, \Sigma_2]\right).$$

※ Generically, $S_{\text{eff}}[\phi_1, \phi_2]$ is not separable into the ϕ_1 sector and the ϕ_2 sector.

cf. Lindblad equation: $\dot{\rho} = -i[H, \rho] + \sum_I \left(L_I \, \rho \, L_I^\dagger - \frac{1}{2}\{L_I^\dagger \, L_I, \rho\}\right)$

SKEFT of the system sector (2/3)

For example, the Langevin dynamics is described by

$$S_{\text{eff}} = \int d^4x \Bigg[\left(-\frac{1}{2}(\partial_\mu \phi_1)^2 - V(\phi_1) \right) - \left(-\frac{1}{2}(\partial_\mu \phi_2)^2 - V(\phi_2) \right)$$

$$-\frac{1}{2}\gamma \left(\phi_1 \dot{\phi}_2 - \dot{\phi}_1 \phi_2 \right) + \frac{i}{2} A (\phi_1 - \phi_2)^2 \Bigg]$$

- The first line is for non-dissipative dynamics.

- The second line is for dissipation and noise, which mix ϕ_1 and ϕ_2.

- The coefficient of the noise term is pure imaginary.

SKEFT of the system sector (3/3)

For intuitive understanding, it is convenient to use the SK basis:

$$\phi = \frac{\phi_1 + \phi_2}{2}, \quad \phi_a = \phi_1 - \phi_2.$$

In this language, the effective action reads

$$S_{\mathrm{eff}} = \int d^4x \left[-\left(-\Box\,\phi + \gamma\dot{\phi} + V'(\phi) \right) \phi_a \, + \, \frac{i}{2} A \phi_a^2 \, + \, \mathcal{O}(\phi_a^3) \right].$$

- The effective action vanishes if we set $\phi_a = 0$, i.e., $\phi_1 = \phi_2$.

- The first term gives the eom w/o the noise.

- The second term capture the noise dynamics.

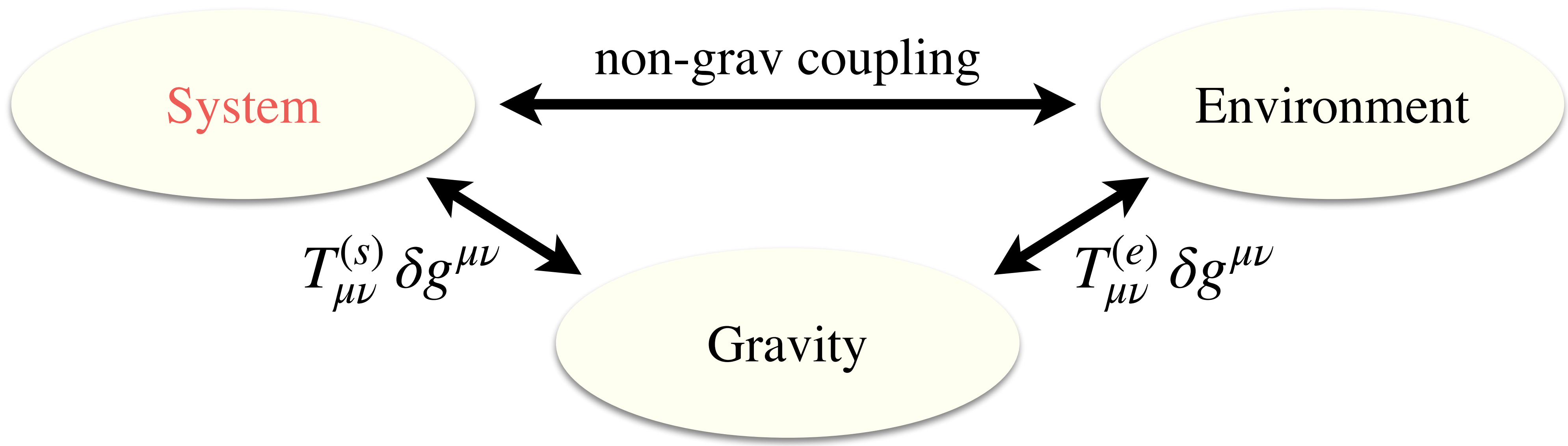
See, e.g., our paper for a review of general properties of SKEFT

such as consequence of microscopic unitarity.

Contents

Now let us incorporate dynamical gravity in this framework.

Gravitational EFT for open systems



Key aspects of dynamical gravity:

- Gravity couples to all the degrees of freedom universally.

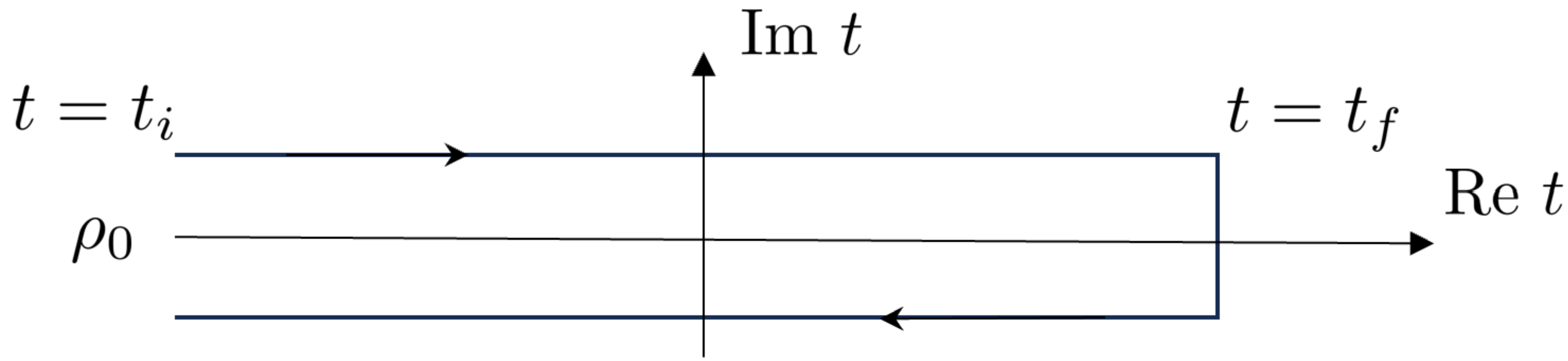
- The effective action has to respect diffeomorphism symmetry.

1) Global symmetry in SK formalism

2) Diffeomorphism symmetry

3) A proposal for the effective action

Global symmetry in the SK formalism (1/2)



In the SK formalism, path-integral variables are doubled.

Accordingly, notion of **symmetry is also doubled**!

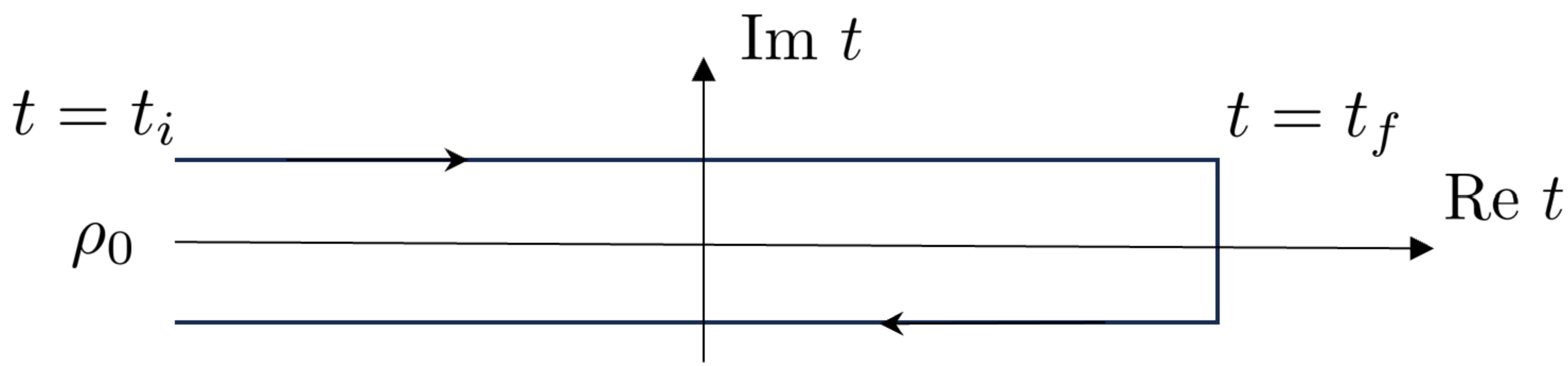
ex. Action of non-dissipative systems: $S_{\mathrm{SK}}[\chi_1, \chi_2] = S[\chi_1] - S[\chi_2]$

- There is no mixing between the $1, 2$ sector.

- If the action $S[\chi]$ enjoys a symmetry G,
  the SK action enjoys the doubled symmetry $G_1 \times G_2$.

- However, boundary conditions at $t = t_f$ break it into a diagonal one.

Global symmetry in the SK formalism (2/2)



In the presence of noise & dissipation, χ_1 and χ_2 are mixed.

Schematically, $S_{\mathrm{SK}}[\chi_1, \chi_2] = S[\chi_1] - S[\chi_2] + S_{\mathrm{diss}}[\chi_1, \chi_2]$.

※ The doubled symmetry $G_1 \times G_2$ is broken into a diagonal one G.

<u>physical intuition</u>

- The diagonal symmetry is associated with the conservation law of the total current including the environment sector current.
- The doubled symmetry $G_1 \times G_2$ is associated with conservation of the system sector current.

ex. Dissipative scalar

Recall the effective action for a dissipative scalar:

$$S_{\mathrm{eff}} = \int d^4x \Bigg[\left(-\frac{1}{2}(\partial_\mu \phi_1)^2 - V(\phi_1) \right) - \left(-\frac{1}{2}(\partial_\mu \phi_2)^2 - V(\phi_2) \right)$$

$$-\frac{1}{2}\gamma \left(\phi_1 \dot{\phi}_2 - \dot{\phi}_1 \phi_2 \right) + \frac{i}{2} A (\phi_1 - \phi_2)^2 \Bigg]$$

- Doubled time-translation: $\phi_i \to \phi_i'(t, \boldsymbol{x}) = \phi_i(t + \epsilon_i, \boldsymbol{x})$ $(i = 1{,}2)$.

- The effective action respects diagonal time-translation $\epsilon_1 = \epsilon_2$, reflecting the energy conservation of the total system.

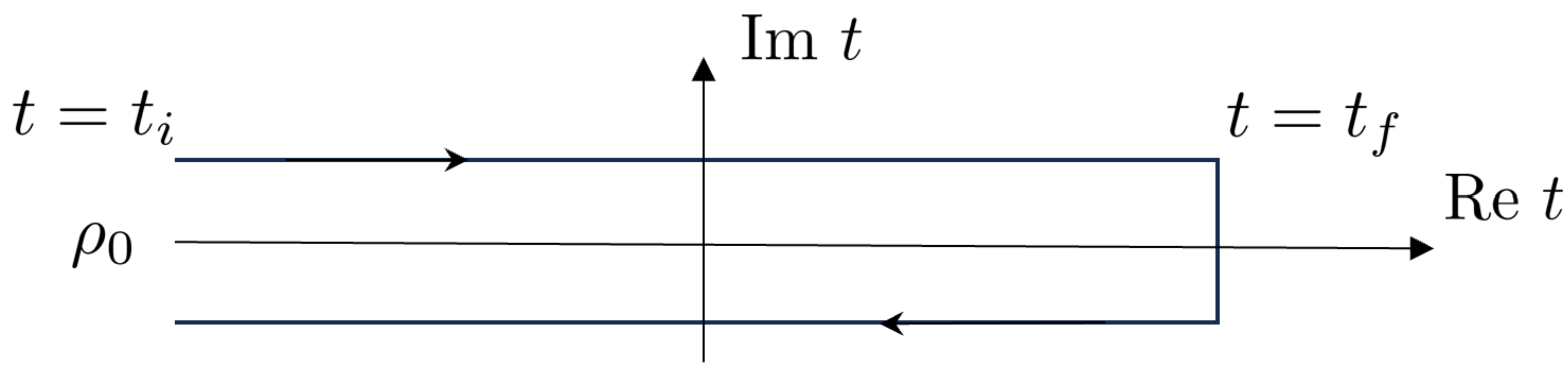
- Off-diagonal time-translation $\epsilon_1 \neq \epsilon_2$ is broken by the second line, due to the mixing of ϕ_1, ϕ_2 that describe noise and dissipation.

1) Global symmetry in SK formalism

2) Diffeomorphism symmetry

3) A proposal for the effective action

Diffeomorphism symmetry



Diffeomorphism symmetry is a local symmetry!

- Doubled diffeomorphism symmetries:

$$\delta\phi_i = \epsilon_i^{\mu}\partial_\mu\phi_1, \quad \delta g_{i,\mu\nu} = \nabla_\mu\epsilon_{i\nu} + \nabla_\nu\epsilon_{i\mu} \;\; (i = 1,2).$$

- The boundary conditions at $t = t_f$ break $\epsilon_1(t_f, \boldsymbol{x}) \neq \epsilon_2(t_f, \boldsymbol{x})$,

but for a generic time t in the bulk, $\epsilon_1(t, \boldsymbol{x}) \neq \epsilon_2(t, \boldsymbol{x})$ is unbroken.

→ **The bulk effective action has to respect the doubled diffs!**

Dissipative scalar revisited

Recall that the Lagrangian of the dissipative scalar contains
local terms like $\phi_1(x)\,\phi_2(x)$ that mix the $1,2$ sectors,
which are an obstruction to realizing off-diagonal diffs $\epsilon_1^\mu \neq \epsilon_2^\mu$
at least within the derivative expansion.

A plausible way to recover the off-diagonal diffs.
is to introduce the Nambu-Goldstone bosons X_i^μ (Stuckelberg trick)
which nonlinearly transform under doubled diffs: $\delta X_i^\mu = -\,\epsilon_i^\mu$.

<u>General lesson</u>

Nambu-Goldstone bosons X_i^μ for doubled diffs are necessary

to make local dissipation / noise compatible with dynamical gravity.

Two illustrative examples for NG bosons X^μ

- Worldline EFT (ex. dissipative tidal response)
- Hydro EFT (a model of the environment bath)

Two illustrative examples for NG bosons X^{μ}

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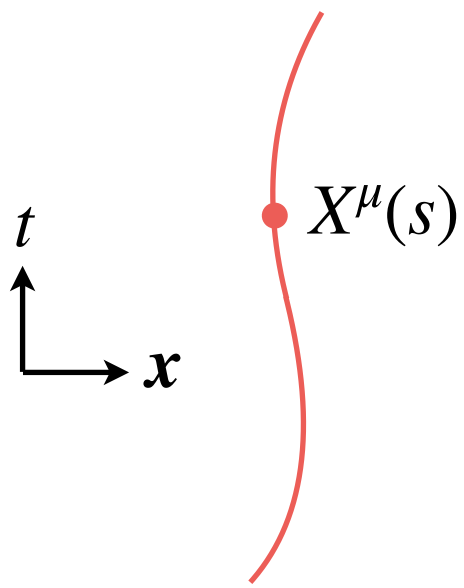
Worldline EFT

Point-particle action



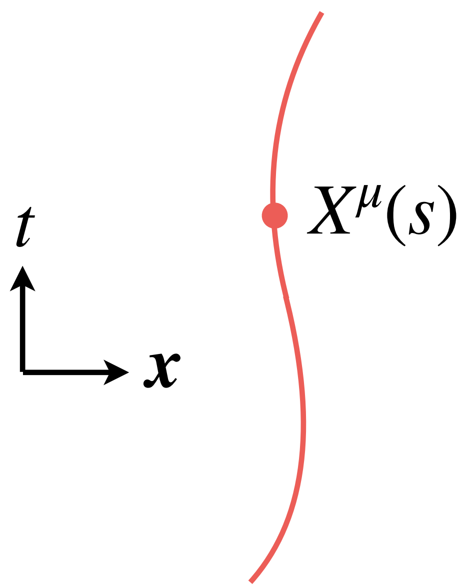
$$S = -m\int ds\sqrt{-g_{\mu\nu}(X)\frac{dX^\mu}{ds}\frac{dX^\nu}{ds}} = -m\int d\tau \quad (\tau : \text{proper time})$$

- $X^\mu(s)$ are spacetime coordinates of the point particle

- $X^\mu(s)$ nonlinearly realize the diffeomorphism symmetry!

Worldline EFT

Point-particle action



$$S = -\,m\int ds\sqrt{-g_{\mu\nu}(X)\frac{dX^\mu}{ds}\frac{dX^\nu}{ds}} = -\,m\int d\tau \quad (\tau : \text{proper time})$$

- $X^\mu(s)$ are spacetime coordinates of the point particle
- $X^\mu(s)$ nonlinearly realize the diffeomorphism symmetry!

Worldline effective action

$$S = -\,m\int d\tau\left[1 + \lambda E_{\mu\nu}^2 + \cdots\right] \quad \left(E_{\mu\nu} := C_{\mu\alpha\nu\beta}u^\alpha u^\beta,\ \ u^\mu := \frac{dX^\mu}{d\tau}\right)$$

- Finite-size effects are captured by higher order corrections.
- The leading order corrections gives the static tidal Love number.

Worldline Schwinger-Keldysh EFT

[in progress w / Ishikawa, Yoshimura]

Point-particle action in the SK formalism

$$S_{\mathrm{pp}} = \frac{m}{2}\int d\tau \left[u^\mu u^\nu g_{a\mu\nu} - 2\frac{Du^\mu}{D\tau}X_{a\mu}\right] + \mathcal{O}(a^3)$$

- $\mathcal{O}(X_a)$ term reproduces geodesic eq. $\dfrac{DX^\mu}{D\tau} = 0.$
- $\mathcal{O}(g_{a\mu\nu})$ term gives a source of Einstein eq.

Covariant time-derivative:

$$\frac{DV^\mu}{D\tau} = \frac{dV^\mu}{d\tau} + \Gamma^\mu_{\nu\rho}u^\nu V^\rho = u^\nu \nabla_\nu V^\mu$$

Worldline Schwinger-Keldysh EFT

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Point-particle action in the SK formalism

$$S_{\rm pp} = \frac{m}{2}\int d\tau\, u^\mu u^\nu \left[g_{a\mu\nu} + \nabla_\mu X_{a\nu} + \nabla_\nu X_{a\mu} \right] + \mathcal{O}(a^3)$$

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- X_a^μ nonlinearly realizes the off-diagonal diffs.

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Worldline SKEFT

$$S = \frac{m}{2}\int d\tau\, u^\mu u^\nu \Big[g_{a\mu\nu} + \nabla_\mu X_{a\nu} + \nabla_\nu X_{a\mu} \Big] \Big[1 + \tilde{\lambda} E^2_{\mu\nu} + \tilde{\nu} E_{\mu\nu} \frac{DE^{\mu\nu}}{D\tau} + \dots \Big] + \mathcal{O}(a^2)$$

- Dissipative tidal effects can appropriately be incorporated.

- Noise effects are also captured by $\mathcal{O}(a^2)$ terms.

- Coupling to elemag fields can be introduced in a similar fashion.

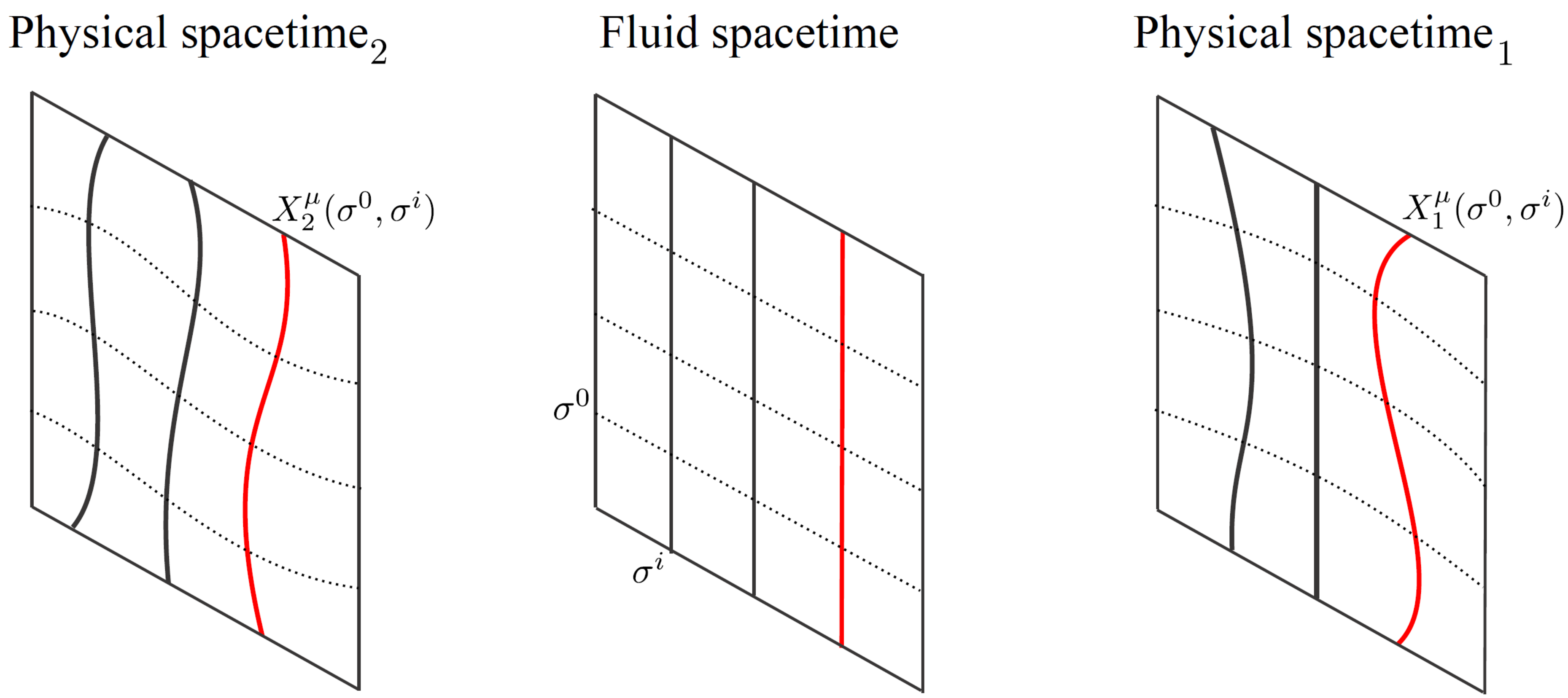
Complementary to existing approaches? Phenomenological application?

Two illustrative examples for NG bosons X^μ

- Worldline EFT (ex. dissipative tidal response)

- Hydro EFT (a model of the environment bath)

HydroEFT (1/2)



[Fig: Glorioso-Liu '18]

Dynamical variables in EFT of dissipative hydrodynamics are spacetime coordinates $X^\mu(\sigma^0, \sigma^i)$ of the fluid element.

※ σ^0 : fluid time, σ^i : label of the fluid elements

※ The SK formalism uses a pair of NG bosons $X_i^\mu(\sigma^0, \sigma^i)$.

※ See, e.g., [Glorioso-Liu '18] for recent developments on HydroEFT.

HydroEFT (2/2)

Schematic form of the SK action for dissipative fluids

$$S_{\text{hydro}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} T^{(\text{hydro})}_{\mu\nu} \mathcal{G}^{\mu\nu}_a + \frac{i}{4} W_{\mu\nu\rho\sigma} \mathcal{G}^{\mu\nu}_a \mathcal{G}^{\rho\sigma}_a \right]$$

※ $\mathcal{G}_{a\mu\nu} = g_{a\mu\nu} + \nabla_\mu X_{a\nu} + \nabla_\nu X_{a\mu}$ is a covariantized noise metric.

※ EOM of X^μ_a guarantees conservation of the stress tensor $T^{(\text{hydro})}_{\mu\nu}$.

※ $T^{(\text{hydro})}_{\mu\nu}$ and $W_{\mu\nu\rho\sigma}$ are constructed from X^μ systematically,

based on the symmetry of fluid spacetime [see, e.g., Glorioso-Liu '18 for details].

HydroEFT (2/2)

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A concrete form of the stress tensor

$$T^{(\text{hydro})}_{\mu\nu} = \rho_0\, u_\mu u_\nu + p_0\,\Delta_{\mu\nu} - 2\eta\left(K_{\mu\nu} - \tfrac{1}{3}\Delta_{\mu\nu}K\right) - \zeta\Delta_{\mu\nu}K.$$

※ $u^\mu \propto \frac{dX^\mu}{d\sigma^0}$: four-velocity of the fluid element.

※ $\Delta_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$: induced metric, $K_{\mu\nu} = \Delta_\mu{}^\rho\nabla_\rho u_\nu$: extrinsic curvature.

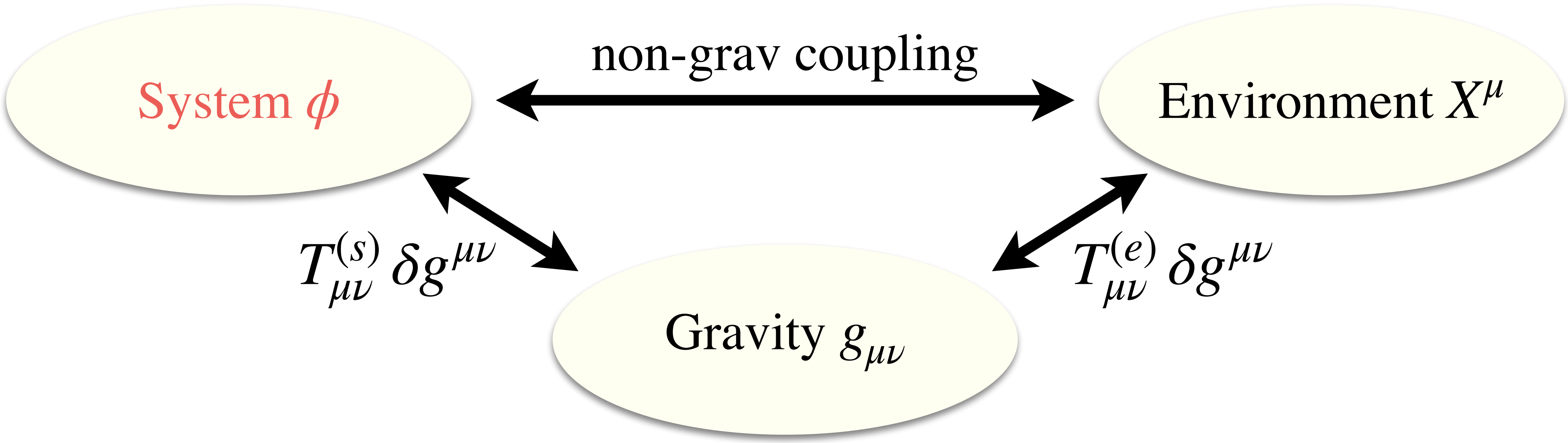
※ ρ_0, p_0, η, ζ are functions of local inverse temperature β.

1) Global symmetry in SK formalism

2) Diffeomorphism symmetry

3) A proposal for the effective action

Gravitational EFT for open systems



Minimal ingredients of the SKEFT :

- The system dof of interests (ex. Dissipative scalar ϕ)
- Dynamical metric $g_{\mu\nu}$ for gravity
- Stuckelberg fields X^μ that nonlinearly realize doubled diffs, which are identified with slow modes of the environment.

EFT is constructed based on nonlinearly realized doubled diffs and consistency requirements of microscopic unitarity etc.

ex. Dissipative scalar coupled to gravity & hydro

The effective action is schematically given by $S_{\rm eff} = S_\phi + S_{\rm grav,\,env}$.

- S_ϕ is a covariantized action for a dissipative scalar:

$$S_\phi = \int d^4x\sqrt{-g}\left[-\left(-\Box\phi + \gamma u^\mu\partial_\mu\phi + V'(\phi)\right)\Phi_a + \frac{i}{2}A\Phi_a^2\right]$$

※ $\Phi_a = \phi_a + X_a^\mu\partial_\mu\phi$ is a doubled diffs invariant combination.

※ Doubled diffs require interactions between system & environment.

※ Higher orders in a-variables are dropped for simplicity.

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※ $\Phi_a = \phi_a + X_a^\mu\partial_\mu\phi$ is a doubled diffs invariant combination.

※ Doubled diffs require interactions between system & environment.

- $S_{\rm grav,\,env}$ is for gravity and the environment sector:

$$S_{\rm grav,\,env} = \int d^4x\sqrt{-g}\left[\frac{1}{2}\Big(-M_{\rm Pl}^2G_{\mu\nu} + T_{\mu\nu}^{(\phi)} + T_{\mu\nu}^{(\rm hydro)}\Big)\,\mathcal{G}_a^{\mu\nu} + \frac{i}{4}W_{\mu\nu\rho\sigma}\mathcal{G}_a^{\mu\nu}\mathcal{G}_a^{\rho\sigma}\right]$$

※ $\mathcal{G}_{a\mu\nu} = g_{a\mu\nu} + \nabla_\mu X_{a\nu} + \nabla_\nu X_{a\mu}$ is a covariantized noise metric.

※ EOM of X_a^μ guarantees conservation of the total stress tensor.

※ Higher orders in a-variables are dropped for simplicity.

Possible application to warm / dissipative inflation

[in progress w / Nishii, Sano]

Warm / dissipative inflation

Warm / dissipative inflation

- Inflaton eom has a local dissipative term, $\ddot{\phi} + (3H + \gamma)\dot{\phi} + V'(\phi) = 0$, which can be used to slow the rolling inflaton.

- In order to induce a local dissipation term, we typically need a large number of dof or strong dynamics in environment, which is an obstruction to model building / pheno of this class of models.

- Our EFT offers a self-consistent framework that describes dissipative inflaton, gravity as well as slow modes of the environment sector.

Q. How large is backreaction from environment bath to inflaton fluctuations?

Comments on decoupling regime

System $\delta\phi$ $\longleftrightarrow$ Environment δX

linear mixing

$\sim \gamma\dot{\bar{\phi}}\ \partial\delta\phi\,\delta X$

kinetic term of δX:

$\sim (\rho + p)_{\rm env}\,(\partial\,\delta X)^2$

amplitude of mixing coupling $\sim \dfrac{\gamma}{H}\sqrt{\dot{\bar{\phi}}^{\,2} / (\rho + p)_{\rm env}}$

- If the environment "kinetic energy" is much larger than the inflaton one,

  slow modes δX in the environment do not affect dynamics of $\delta\phi$.

  ※ Backreaction is negligible if the environment bath is large enough.

- In this regime, however, the main source of background "kinetic energy"

  is the environment, so we need to be careful about ζ vs $\delta\phi$ etc.

※ More quantitative analysis is in progress.

Summary and prospects

Summary

- In the Schwinger-Keldysh EFT, path-integral variables and accordingly would-be symmetries are doubled.

- To couple open systems with a local dissipation / noise term to gravity, it is necessary to introduce Stuckelberg fields for doubled diffs.

- It is convenient to use HydroEFT to model stress tensor of environment, to which energy of the system escapes.

- Under an appropriate scaling assumption, fluctuations in the environment and gravity are negligible, recovering the standard open system EFT.

Prospects

- Our preliminary analysis suggests that a careful analysis on backreaction from the environment bath is required for warm / dissipative inflation. It is important to revisit cosmological perturbations based on our EFT.

- In the context of Black Hole & Gravitational Waves, there are recent active interactions between GR and hep-th (amplitude & EFT) communities. To incorporate irreversible process related to the BH horizon, the SKEFT with gravity will play an important role.

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Thank you!